2 **(a)** **(i)** Define *gravitational potential* at a point.

……………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(ii) Use your answer in **(i)** to explain why gravitational potential near an isolated mass is always negative.

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…………………………………………………………………………………………… [2]

(b) An isolated solid sphere of radius r may be assumed to have its mass M concentrated at its centre. The magnitude of the gravitational potential at the surface of the sphere is ϕ.

On Fig. 2.1, sketch the variation of the gravitational potential with distance d from the centre of the sphere for values from $d = r$ to d = 4r.

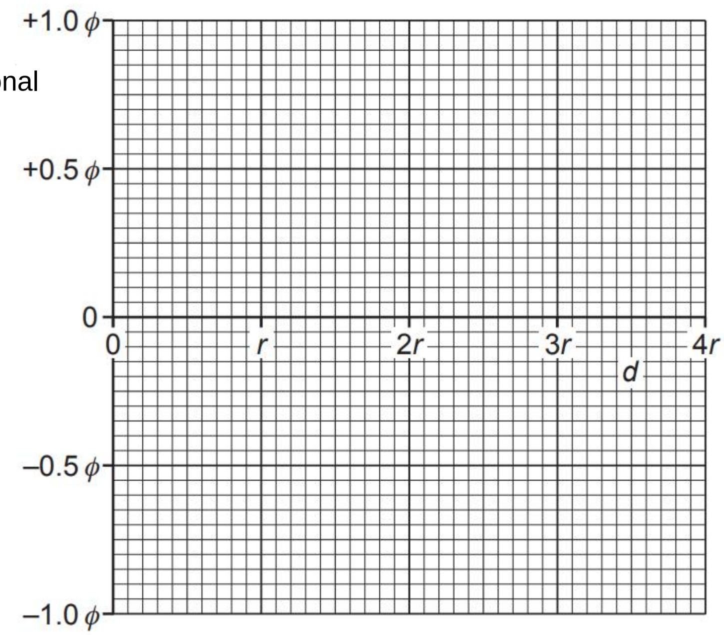

Fig. 2.1

[2]

(c) The sphere in **(b)** is a planet with radius r of 6400 km and mass M of 6.0×10^{24} kg. The planet has no atmosphere.

A spacecraft of mass 8600 kg is to be put into circular orbit about this planet. It orbits at a distance 4r from the centre of the planet.

(i) Show that the speed of the spacecraft at this orbit is 4.0×10^{3} m s^{-1}.

[2]

(ii) The spacecraft then moves to another orbit. Its distance from the centre of the planet changes from 4r to 3r.

1. Calculate the change in gravitational potential energy of the spacecraft.

change = ……….……………….….... J [2]

2. By considering changes in gravitational potential energy and in kinetic energy of the spacecraft or otherwise, determine quantitatively whether the total energy of the spacecraft increases, decreases or remains the same.

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……………………………………………………………………………………….. [2]

3 **(a)** State the first law of thermodynamics